Country Nuclear Power Profiles
2022 Edition

ARMENIA

(Updated 2022)

PREAMBLE AND SUMMARY

This report provides information on the status and development of nuclear power programmes in Armenia, including factors related to the effective planning, decision making and implementation of the nuclear power programme that together lead to safe and economical operations of nuclear power plants.

The CNPP summarizes organizational and industrial aspects of nuclear power programmes and provides information about the relevant legislative, regulatory and international framework in Armenia.

Armenia has one nuclear power reactor unit (Unit 2) in operation that provides about one third of domestic electricity, while its first unit is in permanent shutdown. At Armenian NPP (ANPP, also sometimes referred to as Metsamor NPP), there are lifetime extension activities underway at Unit 2 to extend its operations through 2027. The lifetime extension of Unit 2 is one of the main priorities of the Government of the Republic of Armenia. If safe operation after 2026 is substantiated as a result of relevant studies, the Government of the Republic of Armenia intends to operate Unit 2 at least until 2036. According to forecasts, an additional investment of $150 million will be required. Construction of a new nuclear power plant unit after that term is the main goal of Government of the Republic of Armenia. The presence of nuclear power in the energy portfolio will allow for the diversification of energy resources, a decrease in the dependency on imported natural gas and a decrease in the amount of CO_2 emissions. The above mentioned projects are included in the Armenian Energy Sector Development Strategy Programme (until 2040) which was adopted on January of 2021 by the Government Decision. This Energy Development Strategy Programme stipulates that Armenia envisages construction of a new nuclear unit.

1. COUNTRY ENERGY OVERVIEW

1.1. ENERGY INFORMATION

1.1.1. Energy policy

Prior to 1991, Armenia, as a part of the then Soviet Union (USSR), followed the unified all-union energy policy. At that time, electricity generated by Armenian power plants was connected to the Transcaucasian Energy System. After becoming an independent state, Armenia had to meet open market requirements in all the branches of the energy industry. The energy sector and the nuclear energy sector, in particular, were deeply affected by the economic difficulties during transitions in the market and in need of reorganization and deregulation. Since then, the Armenian energy sector has been modernized and a number of new laws and governmental decisions have been adopted to stabilize the energy sector.

On 7 March 2001, the National Assembly of the Republic of Armenia adopted the Law on Energy of the Republic of Armenia. According to this law, the main principle of the Government's policy on the energy sector is the separation of economic activity, state management and regulation functions. Under this main regulating principle, the rights of consumers and economic interests in the energy sector were balanced. On 25 December 2003, the National Assembly adopted the Law on the Public Services Regulatory Commission, giving the Public Services Regulatory Commission the authority to regulate.

On 1 February 1999, the National Assembly adopted the Law on Safe Use of Nuclear Energy for Peaceful Purposes.

On 21 March 2000, the National Assembly adopted the Law on Amendments and Additions to the Law on Safe Use of Nuclear Energy for Peaceful Purposes. One of the amendments reads: “Those facilities that are important in view of nuclear safety must be constructed and decommissioned by the law”.

On 4 November 2004, the National Assembly adopted the Law on Amendments and Additions to the Law on Safe Use of Nuclear Energy for Peaceful Purposes, according to which the newly constructed nuclear power facilities in Armenia can be owned by different types of entities, radioactive waste (radwaste) and spent nuclear fuel remain state owned, and the operators of nuclear facilities cannot declare bankruptcy.

On 30 September 2013, the National Assembly adopted the Law on Amendments and Additions to the Law on Safe Use of Nuclear Energy for Peaceful Purposes. Those amendments and additions were associated with the accounting and control of nuclear material.

On 16 March 2004, amendments were made to the Law on Licensing, according to which it is necessary to have a licence for the following activities: the design, site selection, construction, operation, decommissioning and the like of nuclear facilities; radwaste storage and disposal; nuclear materials and radwaste processing; transportation and other activities. The rules for obtaining licences for these activities were established by a number of appropriate government decrees.

On 8 December 2005, an amendment was made to the Law on Population Protection in Emergency Situations, according to which, in the event of a nuclear or radiation emergency at a nuclear power plant (NPP), the functions of all involved responsible organizations shall be determined by Government decree. On 22 December 2005, Government Decree No. 2328, National Plan for Population Protection in case of Nuclear and/or Radiation Emergency at the Armenian NPP, was issued. As a result of nuclear and/or radiation emergency exercises at the NPP, a new version of the National Plan for Population Protection in case of Nuclear and/or Radiation Emergency at the Armenian NPP was created and adopted by Government Decree No. 194 on 17 January 2008.

In conclusion, the main goals of energy policy development in the Republic of Armenia are to assure energy independence and energy security by using its own energy sources. Nuclear energy is an independent and low carbon emission energy source.

1.1.2. Estimated available energy

The main sources of energy that are traditionally used in Armenia are natural gas, oil products, nuclear energy, hydropower and coal. Hydropower and a small amount of brown coal are the only domestic sources of energy which are exploited for electricity generation. On 29 December 2016, the Government approved the Hydro Energy Development Concept of Armenia. Armenia has some gas reserves (not exploited) and no oil reserves. The geological forecast projects that some quantity of uranium may exist in Armenia, and in July 2008, a Armenian–Russian joint venture was established for uranium geological exploration and mining. Within the framework of this project, the archival material relevant to uranium mining was collected and analysed. The report Geologic Exploration Activity for 2009–2010, focusing on uranium ore exploration in Armenia, was developed and approved. According to this report, in the spring of 2009 the field work related to uranium ore exploration commenced close to Lernadzor in the province of Syunik, and it was ongoing as of mid-2012. Since it had not met with promising results by the end of 2013, the Armenian–Russian Joint Venture Company was closed.

Energy reserves are shown in greater detail in Table 1. At present, to meet its energy requirements, Armenia imports gas, oil products and nuclear fuel. No new domestic energy sources were discovered in 2020.

TABLE 1. ESTIMATED ENERGY RESERVES

	Estimated available energy sources					
	Fossil fuels			Nuclear	Renewables	
	Solid[1]	Liquid[2]	Gas[3]	Uranium[4]	Hydro[5]	Other renewables[5] (wind)
Total amount in specific units*	—	—	176.0	—	7.0	1.1
Total amount in petajoules (PJ)	—	—	6.0	—	25.0	4.0

Note: Estimated energy reserves (solid and liquid in million tonnes, uranium in metric tonnes, gas in billion m^3, hydro and renewable in TWh per year).

[1] Coal, including lignite: proven recoverable reserves, the tonnage within the proven amount in place that can be recovered in the future under present and expected local economic conditions with existing available technology.

[2] Crude oil and natural gas liquids (oil shale, natural bitumen and extra-heavy oil are not included): proven recoverable reserves, the quantity within the proven amount in place that can be recovered in the future under present and expected local economic conditions with existing available technology.

[3] Natural gas: proven recoverable reserves, the volume within the proven amount in place that can be recovered in the future under present and expected local economic conditions with existing available technology.

[4] Reasonably assured resources (RAR) < USD 130/kgU.

[5] Hydropower and renewable technically exploitable capability, the amount of the gross theoretical capability that can be exploited within the limits of current technology.

* Sources: World Energy Resources: Survey 2013, WEC 2013 and Uranium 2018: Resources, Production and Demand (“Red Book”), IAEA Energy and Economic Databank; Country Information.

1.1.3. Energy Consumption Statistics

Primary energy sources, in petajoules (PJ), are summarized in Table 2. To meet its energy requirements, Armenia imports gas, oil products and nuclear fuel.

TABLE 2. ENERGY CONSUMPTION

Final Energy consumption [PJ]	2000	2005	2010	2015	2020	Compound annual growth rate 2000–2020 (%)
Total	46	72	76	90	109	4.38
Coal, Lignate and Peat	0	0	0	0	0	0.00
Oil	12	16	16	13	22	3.10
Natural gas	18	39	43	53	62	6.24
Bioenergy and Waste	1	0	0	4	3	9.95
Electricity	13	15	17	19	21	2.47
Heat	2	2	0	0	0	-8.83

*Latest available data, please note that compound annual growth rate may not be representative of actual average growth.

**Total energy derived from primary and secondary generation sources. Figures do not reflect potential heat output that may result from electricity co-generation.

—: data not available.

Source(s): United Nations Statistical Division, OECD/IEA and IAEA RDS-1

http://www.mtad.am/u_files/file/energy/Energy%20Balance%202018-merged.pdf

1.2. THE ELECTRICITY SYSTEM

1.2.1. Electricity system and decision making process

Hydropower is based on Armenia's water resources, including Lake Sevan, one of the largest highland freshwater lakes in the world (1900 m above sea level), and the rivers Arax, Arpa, Hrazdan, Debet and Vorotan. The hydropower plants (HPPs) of the Sevan–Hrazdan cascade are operating at a low level, because, after the intense use of lake water, the Government of Armenia decided to reduce releases from Lake Sevan to restore its potential. Water from the lake can be taken only for irrigation needs. Two HPP cascades and small HPPs have a total installed capacity of 1334 MW, of which:

- Sevan–Hrazdan HPP cascade has an installed capacity of 550 MW.
- Vorotan HPP cascade has an installed capacity of 400 MW.
- Small HPPs have an installed capacity that increased up to 384.4MW in 2021.

Electricity production by small HPPs in 2021 was about 0. 79 TWh. At the same time, Armenia still has an unused hydraulic potential (both small and big rivers) of about 500 MW (or 2,0 billion kWh of electric energy), with development being economically reasonable.

The thermal power plants (TPP) have an installed capacity of 2560 MW, of which:

- Hrazdan TPP had an installed capacity of 1100 MW: 4 units of K - 200 condensation turbines, 2 units of T - 100 and 2 units of T - 50 MW heating turbines. Only two units of 200 MW are now in operation because the operational lifetimes for the others have already expired. Operational capacity of these two units are 370 MW. In April 2012 the new gas and steam turbine units with a summary capacity of about 440 MW (Hrazdan - 5) were put in operation at the Hrazdan TPP site. Now the total capacity of Hrazdan TPP and Hrazdan - 5 is 370 MW + 440 MW = 810 MW. In 2021 Hrazdan – 5 unit was not in operation.
- Yerevan TPP had an installed capacity of 550 MW, including 2 × 150 MW condensation turbines and 5 × 50 MW heating turbines. The operational lifetime for all the units has already expired. In April 2010, the gas turbine combined cycle unit of Yerevan TPP, with a capacity of about 220 MW electrical and 30 MW thermal, was put into operation. On 4 March 2019, the Ministry of Energy Infrastructure and Natural Resources signed an agreement with Renco regarding construction of a new unit with 250 MW capacity at the Yerevan TPP. In September 2021, unit with 250 MW capacity at the Yerevan TPP was put in operation. The total capacity of Yerevan TPP is now 470MW.
- Vanadzor TPP had an installed capacity of 94 MW, with different capacity heating turbines. None of them are now in operation because the operational lifetime for all the units has already expired.

The total capacity of TPP is now 1280 (810 MW, Hrazdan TPP and Hrazdan – 5 + 470 MW) (total of Yerevan TPP).

The nuclear power plant (ANPP) has a designed capacity of 815 MW, of which Unit 2 with 407.5 MW is in operation. Nuclear energy played a crucial role during the period of recovery from the economic crisis. Unit 1 is not operating, and Unit 2 was recommissioned in 1995, after 6.5 years of outage. The fuel is supplied by the Russian Federation. Within the Unit 2 lifetime extension project the turbines have been modernized. Due to increased efficiency of turbines the total capacity of Unit 2 reached 451 MW for the same thermal capacity of nuclear reactor. Now ANPP's Unit 2 is operated with 415 MW capacity taking into account the license permission to operate with 92 % of thermal capacity. In 2021 turbogenerators and cooling towers of Unit 2 were modernized. As a result, the unit's capacity will increase up to 448 MW. Armenian NPP submitted appropriate documents to ANRA and has received its license to operate with 98% thermal capacity.

The high voltage transmission network of Armenia consists of 220 and 110 kV lines. There are 14 substations of 220 kV and 119 substations of 110 kV. The capacity of the existing high voltage network is considered sufficient for the current and forecasted loads. The high voltage transmission network has interconnections with all neighbouring countries: Azerbaijan: 330, 220 and 110 kV (not in operation); Georgia: 220 and 110 kV; Islamic Republic of Iran: 2 × 220 kV; and Türkiye: 220 kV (not in operation). The high voltage lines Armenia–Islamic Republic of Iran and Armenia–Georgia of 400 kV are currently under construction.

Natural gas is the most important primary energy source, and it is imported primarily from the Russian Federation. The designed capacity of the high pressure gas transportation network of Armenia is 17 billion m^3/y. In 1980, the maximum demand for natural gas in Armenia was above 5–6 billion m^3/y. Five main gas pipelines were built, which ensured gas delivery from three sides: Georgia, North Azerbaijan and West Azerbaijan. Today, only the Georgian pipeline is operating. In 2021, the natural gas demand was 2.794 billion m^3, where 2.449 billion m^3 was imported from Russian Federation and 0.345 billion m^3 from Islamic Republic of Iran. But the expected demand will be 5.5–6.2 billion m^3/y, depending on the ANPP's status (shut down or in operation). The gas pipeline Armenia–Islamic Republic of Iran, which is now fully constructed and has been in operation since spring 2009, has a capacity of 2.3 billion m^3. There are underground storage facilities for natural gas with a maximal gas storage volume of 180 million m^3. Currently, the available gas storage volume is 130 million m^3. Gas distribution in Armenia is performed through high, medium and low pressure distribution networks. In 2021, the energy sector used 0. 504 billion m^3 gas, gas demand by population was 0.773 billion m^3, gas stations for cars used 0.505 billion m^3.

Oil products are imported from foreign countries and mostly utilized for the transportation and industry sectors. In 2021 630 Mtoe oil products were imported. During the past several years, mazut (a low quality heavy fuel oil) has barely been imported into Armenia.

Other energy sources (geothermal, wind, solar and waste burning) are under consideration. Armenia has considerable potential for geothermal energy, but a programme has to be developed to explore its geothermal resources and to carry out drilling activities.

According to the Wind Energy Resource Atlas of Armenia, economically reasonable wind power potential is estimated at 450 MW total installed capacity and at electric power output of 1.26 billion kWh/y. The most promising locations are the Zod (Sotk) Pass, the Bazum Range, the Pushkin Pass, the Qarakhach Pass, the Jajur Pass, the Geghama Range, the Sevan Pass, the Aparan region, the Sisian–Goris hills and the Meghri area, where the wind average velocity reaches 7 m/s. In December 2005, the first wind power plant was put into operation in the Pushkin Pass (the Vanadzor region) with the installed capacity of 2.6 MW. The total capacity of the site is estimated to be 20 MW. On 01 January 2022, the total capacity of wind power plants reached 4.23 MW and in 2021 total electricity production was 4.02 GWh. Investigations are being carried out for the construction of wind power plants at other sites. The Spanish Acciona company informed that it wanted to construct wind power plants in Armenia with a capacity of 202 MW (with 37 turbines). Project investment is $ 276 million. This wind power plant will be put in operation in 2025.

Armenia has significant solar energy potential. The average annual amount of solar energy flow per square meter of horizontal surface is about 1720 kWh (the European average is 1000 kWh). One fourth of the country's territory is endowed with solar energy resources of 1850 kWh/m^2/y. In January 2022 37 companies received license for construction of solar power plants with total capacity of 214.7 MW. At the beginning of 2022 the total capacity of solar plants reached 50.75 MW. In 2021 the total electricity production by solar plants was 100.0 GWh.

According to the Armenian Energy Sector Development Strategy Programme (until 2040) solar power plants with a total installed capacity of up to 1000 MW will be built in Armenia by 2030.

The financing agreement for the development of the largest utility-scale solar power plant in Armenia is done, "Ameria Bank" will support the project implementation by financing VAT needs. Fotowatio Renewable Ventures (FRV), part of Abdul Latif Jameel Energy, will be supported by the International Finance Corporation (IFC), a member of the World Bank Group, the European Bank for Reconstruction and Development (EBRD) and the European Union (EU). They will extend up to USD 38.4 million financing that is structured in different tranches of debt. The largest utility-scale solar power plant in Armenia will be located in Mets Masrik municipality, in the Gegharkunik region. The 55 MW solar power plant is the first-of-its-kind in the country, whereas the Armenian government has allocated 32.6591 ha of area for this project with its decision 326-L dated on 29 March 2019. "Masrik -1" solar power plant will be put in operation in 2023 with a capacity of 55 MW and will generate more than 128 GWh of electricity per year at a competitive tariff of $ 41.9 per MWh. The electricity will be sold under a power purchase agreement to Electric Networks of Armenia, the utility company responsible for distribution of electricity in the country. The plant will produce enough energy to supply more than 20,000 Armenian households and is projected to avoid the release of over 40,000 tons of carbon emissions annually, according to a bank press release.Cabinet of Ministers of the Republic of Armenia approved the investment program of the Masdar company from Abu Dhabi, United Arab Emirates. The company intends to build solar and wind power stations in Armenia, as well as solar "floating" stations with a total installed capacity of 400 MW. The project itself is planned to be implemented in two stages. During the first stage—until December 31, 2023—it is planned to build solar power plants with a capacity of 200 MW, to develop required infrastructures and to connect the plants to the grid and test them. The proposed tariff for the first stage will be 2.99 cents per kWh, which is 30% less than the lowest tariff in the country. Goals of the second stage, ending by December 31, 2024, are to build another 200 stations with a total capacity of 200 MW, but already with a slightly lower tariff of 2.95 cents. With the support of the European Bank for Reconstruction and Development, the construction programme for an industrial scale solar power plant with 200 MW capacity is also being prepared for the administrative territory of Aragatsotn Province — Dashtadem site.

One biomass plant is operational in Armenia that has a capacity of 0.84 MW and produced 2.32 GWh electricity in 2021. In 2021, installed capacity of all renewable sources was 58.85 MW and production was about 106.6 million kWh of electrical energy.

1.2.2. Structure of electric power sector

The Ministry of Energy and Fuel of Armenia was established in 1992. In May 2008, the Ministry of Energy and Fuel was rebranded as the Ministry of Energy and Natural Resources, and in 2016 as the Ministry of Energy Infrastructure and Natural Resources. On 8 May 2019 the Ministry of Energy Infrastructure and Natural Resources became a part of the Ministry of Territorial Administration and Infrastructure. This Ministry holds the responsibilities of the of energy sector, energy infrastructure and natural resources to ensure development and implementation of state policy on the use of nuclear energy for peaceful purposes, the energy independence of Armenia, and the efficient use of alternative energy sources and domestic energy resources. It is responsible for defining policy for development of the energy sector and infrastructure of Armenia.

The duties of the Committee on Nuclear Safety Regulation of the Republic of Armenia (ANRA) are to perform the nuclear energy regulation and supervision of nuclear powered objects, issue licences and control the fulfilment of licence requirements. Its main objective is to secure the protection of the population, the personnel involved in the nuclear industry, and the environment. On 06 March 2020, the National Assembly adopted the Law on Amendments to the Law on Safe Use of Nuclear Energy for Peaceful Purposes, under which the Committee on Nuclear Safety Regulation of the Republic of Armenia now is authorized to conclude international agreements and adopts rules, requirements, procedures, conditions and lists for regulation of activities in the field of nuclear energy, including: nuclear safety, radiation safety and protection, radioactive waste management, transportation of nuclear and radioactive materials, physical protection of nuclear energy facilities and nuclear and radioactive materials, emergency preparedness and response.

The Public Services Regulatory Commission of Armenia is responsible for antimonopoly regulation. The key functions of antimonopoly regulation are tariff regulation and licensing of entities in the energy sector.

The operator of the electric energy network is responsible for dispatching activity, and the Settlement Centre is in charge of calculating the wholesale trade of electric energy. It also approves the balance between the participants of the trade.

The structure of management of the energy sector in Armenia is shown in Fig. 1.

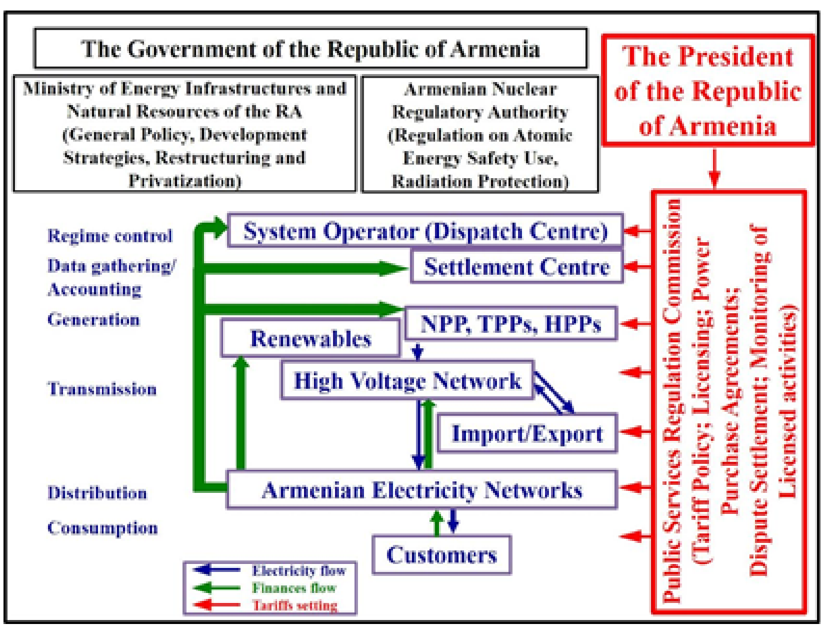

FIG. 1. Structure of management of the energy sector.

The Government dedicates special attention to restructuring the electricity sector, with a number of energy laws that were adopted to achieve its target. A programme for improvement of metering, billing and collection of payments for electricity, heat and natural gas was implemented, coupled with the conversion of the accounting system to international norms and standards and annual auditing of the company's financial reports by independent auditors. In addition, a programme was implemented to organize collections through banks. Despite limited economic difficulties, the Government gives priority to budget payments for the electricity provided to budget organizations, as well as compensation for the electricity consumed by irrigation, drinking water, industry and electrical transport companies.

The implementation of a stabilization policy, with the crucial role of restarting the ANPP, allowed the country to overcome the electric energy crisis of the post-Soviet period. Now, Armenia is covering its electricity demand completely and even exports electric energy to neighbouring countries. In the near future, however, additional energy sources may be required as living standards rise.

For Armenia, it is critical to be involved in the regional power market, which is currently in the process of formation and which foresees potential establishment of a circular power system of Black Sea countries, ideally as well as the creation of north–south parallel operation relations.

1.2.3. Main indicators

In 2019 the total capacity of the electric energy generating plants in Armenia was 4.19 GW(e). The results of asset revaluation show that the sector's main asset resources have already expired. The equipment is worn out and requires a major overhaul — 30% of installed capacities are already over 30 years old.

On 1 January 2022 the total capacity of the electric energy generating plants in Armenia was 3.12 GW(e). In 2021, electricity NET production was 7.39 billion kW?h. Table 3 shows the historical statistics of electricity production and its distribution by plant types; Table 4 shows the energy related ratios. In 2021, the electricity consumption was around 2096 kW?h/capita in Armenia. The electricity consumption of the capital city Yerevan is about 35% of the total electricity consumption in Armenia.

TABLE 3. ELECTRICITY PRODUCTION

Electricity production (GWh)	**2000**	**2005**	**2010**	**2015**	**2020**	**Compound annual growth rate 2000–2020 (%)**
Total	5 958	6 317	6 491	7 799	7 796	1.35
Coal, Lignate and Peat	0	0	0	0	0	0.00
Oil	0	0	0	0	0	0.00
Natural gas	2 692	1 828	1 438	2 801	3 166	0.81
Bioenergy and Waste	0	0	0	0	0	0.00
Hydro	1 261	1 773	2 556	2 206	1 778	1.73
Nuclear	2 005	2 716	2 490	2 788	2 756	1.60
Wind	0	0	7	3	2	0.00
Solar	0	0	0	1	94	0.00

*Latest available data, please note that compound annual growth rate may not be representative of actual average growth.

**Electricity transmission losses are not deducted.

—: data not available.

Source: United Nations Statistical Division, OECD/IEA and IAEA RDS-1

TABLE 4. ENERGY RELATED RATIOS

Final Energy consumption [PJ]	**2000**	**2005**	**2010**	**2015**	**2020**	**2021***
Nulcear/total electricity(%)	33	42.7	39.4	34.5	34.5	25.3

*Latest available data.

Source: RDS-1 and RDS-2

—: data not available.

2. NUCLEAR POWER SITUATION

2.1. HISTORICAL DEVELOPMENT AND CURRENT ORGANIZATIONAL

STRUCTURE

2.1.1. Overview

In September 1966, a decision to construct a nuclear power plant in Armenia was made by the former USSR Council of Ministers and the appropriate decree was issued. In 1968, the Armenian Branch of the Electrosetproject Institute completed the prefeasibility study for constructing the ANPP under the project titled The Scheme of NPP Contribution to the Power Grid. The document included a schedule to commission Unit 1 in 1973 and Unit 2 in 1974.

The technical specification to design the ANPP was developed by Teploelectroproject in 1968 and was approved in August 1969 under Decree No. 1624 R.C. of the former USSR Ministry of Energy.

More than 20 potential sites were considered for the ANPP construction, and finally a site was selected in the western part of the Ararat valley, 16 km from the Turkish border, 10 km to the northeast of the regional centre Hoktemberyan (Armavir), and 28 km (to the west) from Yerevan.

In accordance with that specification, the capacity of the ANPP (first stage of construction) included WWER-440 type reactors with a total of 815.0 MW capacity (407.5 MW for each unit). The ANPP Design Lifetime was specified to be 30 years.

Initial comprehensive studies and analyses showed that the seismic conditions of the ANPP site were characterized at an intensity level of eight points according to the Medvedev–Sponheuer–Karnik (MSK)-64 scale. It was the first nuclear power plant in the former USSR intended to be constructed in a region of high seismicity.

The specific nature of the ANPP site — its seismicity — caused significant changes in the design of WWER-440/230, not only in construction, but also in the design of the reactor facility as a whole; as a result, the reactor was assigned the new identification of V-270. The design of the reactor was based on Units 3 and 4 of the Novovoronezh NPP.

The reactor building, auxiliary building, ventilation stack, as well as the buildings and structures containing equipment and instrumentation of safety systems or safety related on-line systems and communications connecting these structures were assigned a category of high importance. They were designed to have one point more seismic resistance than the overall ANPP site.

The ANPP was commissioned in 1976, achieving initial criticality for Unit 1 on 22 December 1976 and for Unit 2 on 5 January 1980. The units were put into commercial operation on 6 October 1977 and 3 May 1980, respectively.

In 1981, the technical-economic background was developed for further expansion of the ANPP (the second stage of the plant), taking into consideration the central heating needs of Yerevan. The technical-economic background was approved and coordinated with all the relevant organizations. In 1985, the Gorki Department of the Atomteploelectroproject Institute prepared the following project, titled Armenian NPP: Its Expansion, Taking into Consideration the Central Heating in Yerevan City. Excavation work began and the foundation pits for two new units (Unit 3, Unit 4) were dug. However, following the 1986 accident at the Chornobyl nuclear power plant, the Government made a decision to refuse further expansion of the ANPP and construction was stopped as a result.

After the 1988 earthquake, although the ANPP was not damaged, the Council of Ministers of the former USSR decreed that the ANPP be shut down as a precautionary measure. Unit 1 was shut down on 25 February 1989 and Unit 2 on 18 March 1989. The units were not decommissioned but kept in a prolonged shutdown.

In April 1993, the Government of Armenia decided to restart Unit 2 of the ANPP in order to meet energy needs while tackling an economic crisis, taking into account the lack of national energy resources. Following 6.5 years of outage during its shutdown, with the technical and financial help of the Russian Federation, Unit 2 of the ANPP was restarted on 5 November 1995. Unit 1 has remained shut down, however.

Following the decision of the Government on 27 March 2014 to extend operations at ANPP Unit 2, work for the ANPP Unit 2 Design Lifetime Extension was launched. After the work began, and when Design Lifetime Extension activities will be finalized relevant documents will be submitted to the Committee on Nuclear Safety Regulation of the Republic of Armenia to receive a licence for extension of the operational lifetime.

According to the Armenian Energy Sector Development Strategy Program (until 2040), the extension of the operation lifetime of the Armenian NPP’s Unit 2 until 2036 is being considered, for which relevant works are currently being carried out.

In 2020 projects for modernization of equipment of Unit 2 were implemented as a part of life extension activities. Turbine generators have been improved, as a result of which the unit's capacity increased. In 2021 the installed gross capacity reached 448 MW instead of 407.5 MW. As a result of Regulatory Body restrictions of the license the Unit 2 is limited to 92% of the installed thermal capacity.

In 2021 there is a plan to reconstruct cooling towers. Consequently the average annual values of the cooling water temperature will decrease, which in its turn will lead to an increased capacity of Unit 2. Other modernization activities are also considered to increase its capacity. Following the ANPP Unit 2 restart, 58.21 billion kWh of electric energy had been generated by 01 January 2021, keeping to the load schedule of the Armenian power system.

2.1.2. Current organizational structure

Several organizations, institutions and state bodies are currently involved in activity related to the operation of the ANPP.

According to Governmental Decree No. 98 of 4 April 1996, the Closed Joint Stock Company (CJSC) Armenian NPP was created and authorized to act as operator of the nuclear power plant. For other purposes, such as liability to foreign countries, the State is assumed to be the operator.

In Armenia, the ANPP is under state ownership according to the Law on Safe Use of Nuclear Energy for Peaceful Purposes. There is no change in organizational part of ANPP.

ANRA was established in 1993 to be the regulatory body in the area of nuclear and radiation safety, to perform inspection activities and to issue the licences for the appropriate applications (see more detailed information in Section 3.1).

During the period of preparation for the ANPP restart (1993–1996), the Armatomenergo was established under the Ministry of Energy and Fuel. Armatomenergo was authorized with the functions of the operator of the ANPP. On 4 April 1996, by Government Decree No. 98, the operation of the Armatomenergo ceased, and the CJSC Armenian NPP was given the function of operator.

The Armatom Institute was established in 1973. Having provided engineering support to the ANPP, the institute undertakes work such as implementing diagnostic systems, implementing the safety parameters display system and developing compact and multifunctional simulators within the ANPP. Armatom is participating in the development of the following reports: deterministic analysis of ANPP Unit 2 and probabilistic safety analysis of ANPP Unit 2.

CJSC Atomservice was established in 1987. The company took an active part in adjusting the plant systems and implementing test programmes during the period of preparation for the ANPP Unit 2 restart. To date, it continues to perform the same activity.

The CJSC Atomenergoseismoproject was established in 1983. It completed investigations of seismic conditions at the plant for final resolution of all the issues relevant to the plant restart, including potential extensions to its operation. One of the major results of the investigations was that the ANPP was erected on a whole (non-destructed) basalt block; that is, the absence of a tectonically active break under the ANPP site was determined. At present, the CJSC Atomenergoseismoproject is part of CJSC Scientific Research Institute of Energy.

There are several organizations that oversee construction, repair, mounting and other related activities in regard to the operation of the ANPP.

In Armenia, the All-Armenian Atomic Power Engineers Association was established. The founders of the association are specialists from such organizations as the Ministry of Energy Infrastructure and Natural Resources, ANRA, ANPP, the National Polytechnic University of Armenia (NPUA), Yerevan State University (YSU) and other nuclear power specialists. The association’s main objectives are as follows:

- To promote development of scientific ideas in nuclear engineering;
- To support promotion and further development of nuclear energy;
- To conduct research and tests in the field of atomic energy, according to established procedures;
- To organize public discussions addressing issues relevant to nuclear energy;
- To ensure the promotion of nuclear energy by:
    - Publishing articles, magazines, books, dictionaries and reference books;
    - Organizing scientific seminars;
    - Creating radio programmes, documentary and scientific films and videos devoted to nuclear energy;
    - Creating computer training and demonstration programmes.

The Armenian Union of Veterans of Atomic Energy was established. The Union’s main objectives are:

- To unite the veterans/retired professionals of the nuclear energy field in the Republic of Armenia,
- To use their experience and knowledge in the training of young specialists in the frames of knowledge transfer,
- To communicate with the public the topics related to nuclear energy safety and its developments.

2.2. NUCLEAR POWER PLANTS: OVERVIEW

2.2.1. Status and performance of nuclear power plants

The ANPP consists of two WWER-440 (V-270) type nuclear power units that were designed and constructed by organizations of the former USSR under the supervision of the Ministry of Energy and Electrification. The design of the first stage of the plant was developed in 1969–1970. The chief scientific supervisor was the Kurchatov Institute of Atomic Energy in Moscow (now the National Research Center (NRC) Kurchatov Institute). The chief design organization was Teploelectroproject, Gorki, which evolved to be known as the Atomenergoproekt Nizhny Novgorod Engineering Company (NIAEP). The main reactor construction organization was the Experimental Design Bureau Gidropress, Podolsk. The Izhora Factory in Leningrad manufactured the reactors and systems. The turbines were manufactured by the Kharkov Turbine Plant. The electric generators were supplied by the Electrosila plant of Leningrad. Finally, the building construction work was performed by Gidroenergostroy, Yerevan.

Unit 1 has not operated since 1989 and was placed into permanent shutdown. Since its restart in 1995, Unit 2 of the ANPP has been in operation with an installed gross capacity of 407.5 MW.

All the nuclear fuel necessary for operation of ANPP was delivered in the past and is currently being delivered by the TVEL concern of the Russian Federation.

Table 5 shows the status and some other indicators of the nuclear power units of the ANPP.

TABLE 5. STATUS AND PERFORMANCE OF NUCLEAR POWER PLANTS

Reactor Unit	Type	Net Capacity [MW(e)]	Status	Operator	Reactor Supplier	Construction Date	First Criticality Date	First Grid Date	Commercial Date	Shutdown Date	UCF for 2021
ARMENIAN-2	PWR	448	Operational	ANPPCJSC	FAEA	7/1/1975	1/1/1980	1/5/1980	5/3/1980		51.5
ARMENIAN-1	PWR	376	Permanent Shutdown	ANPPCJSC	FAEA	7/1/1969	12/15/1976	12/22/1976	10/6/1977	2/25/1989	

Data source: IAEA - Power Reactor Information System (PRIS).

Note: Table is completely generated from PRIS data to reflect the latest available information and may be more up to date than the text of the report.

In 2015, the ANPP generated 2.79 billion kWh — the maximum amount generated since the ANPP restart.

The main organizations and institutions regarding nuclear energy in Armenia include the Ministry of Territorial Administration and Infrastructure, the Nuclear Safety Regulation State Committee (still referred to as ANRA), CJSC ANPP, CJSC Armatom, CJSC Atomservice and CJSC Atomenergoseismoproject. Some technical support was provided by organizations of the Russian Federation (e.g. Experimental Design Bureau Gidropress — main reactor designer, NIAEP — main NPP designer, NRC Kurchatov Institute — scientific management, among others).

In 1995, Unit 2 of the ANPP had five emergency events at level 0 (below scale, deviation) on the International Nuclear and Radiological Event Scale (INES). In 1996, 8 emergency events occurred at the ANPP, including one at level 1 (anomaly) and seven at level 0 on the INES scale. In 1997, five emergency events occurred at the ANPP, including two at level 1 and three at level 0 on the INES scale. In 1998, seven emergency events occurred at the ANPP, including two at level 2, one at level 1 and four at level 0 on the INES scale. In 1999, one emergency shutdown and one event at level 1 occurred. In 2000, there were three events reported: one event was rated level 1 and two events were rated level 0. In 2001, eight emergency events occurred at the ANPP, including three at level 1 and five at level 0 on the INES scale. In 2002, eight emergency events at level 0 on the INES scale occurred at the ANPP. There were two emergency shutdowns that year. In 2003, there were two emergency events, one at level 0 and one at level 1 on the INES scale. There was one emergency shutdown in 2003. In 2004, there were two emergency events at level 1 on the INES scale. In 2005, 2006 and 2007 no emergency events on the INES scale occurred. In 2008, during the operation of Unit 2 of the ANPP, one event at level 1 was registered, and the reactor was scrammed, which was caused by an accident in the grid. In 2009, there were five recorded events in the plant operation; four events were classified as INES level 0 and one event was classified as a safety significant level 1 event on the INES scale. In 2010, there were seven recorded events in the plant operation and all events were classified as INES level 0. In 2011, there were four recorded events at the plant, and all events were classified as INES level 0. In 2012, during the operation of Unit 2 of the ANPP, two events at level 0 were registered. The reactor was not scrammed. In 2013, there were five recorded events in the plant operation; four events were classified as INES level 0 and one event as INES level 1, and the reactor was manually scrammed. In 2014, there were five recorded events in the plant operation, and five events were classified as INES level 0. In 2015, there were four recorded events and all events were classified as INES level 0. In 2016–2021, during the operation of Unit 2 of the ANPP, 0 events were registered.

On 17 September 2021, the Armenian NPP celebrated the 55th anniversary of the decision to build the first nuclear power plant in Armenia by the Ministry of former USSR.

2.2.2. Plant upgrading, plant life management and licence renewals

The issues of the ANPP safety upgrade are of great importance for the Armenian Ministry of Energy Infrastructure and Natural Resources, and safety is a primary consideration for the Armenian Government. After numerous consultations with experts from the United States of America, countries across Western Europe and the Russian Federation, including assistance from experts from the IAEA, Armenian specialists developed a new programme for the ANPP safety upgrade: the list of safety upgrade activities for the period of 2009–2016 of Unit 2 of the Armenian NPP. The safety upgrade process was previously permanently implemented at the ANPP and is being realized according to the provisions of that programme. Since the restart of the ANPP and as of 1 January 2021, a lot of safety upgrade activities and safety improvement measures (modifications according to technical decisions, improving the safety and reliability of NPP equipment and systems) have been completed, so the plant can withstand emergency situations and minimize potential failures.

On the basis of the Government Session No. 50 Protocol Decision, Point 14, On Approval of the Concept for Ensuring Energy Security in the Republic of Armenia (on 22 December 2011), ANPP Unit 2 lifetime extension activities began and several governmental decisions were adopted.

On 19 April 2012, Government Decision No. 461-N, On Lifetime Extension of Unit 2 of Armenian NPP, was adopted. According to this decision, the Minister of Energy Infrastructure and Natural Resources was assigned to develop the programme on extending the lifetime of ANPP Unit 2 and to estimate the amount of financial resources needed to implement it.

On 23 August 2013, Government Decision No. 1085-N, Requirements for the Lifetime Extension Regarding the Operation of ANPP Unit 2, was adopted.

On 27 March 2014, the Government Session No. 12 Protocol Decision, Point 11, Programme for the Design Lifetime Extension of Unit 2 of Armenian NPP, was adopted. The Government assigned the Minister of Finance to sign the intergovernmental agreement between Armenia and the Russian Federation on credit resources for the implementation of the programme developed under the first point of the decision by 1 May 2014.

On 3 May 2018, Government Decision No. 548-N, Investment Plan and Regarding Implementation Measures of Design Lifetime Extension of Unit 2 of Armenian NPP, was adopted.

According to the Armenian new energy strategy, the current investment programme to extend the operating life of Unit 2 will be completed by 2023.

At the first preparatory stage, a tremendous amount of work was carried out to survey the main equipment. In total, about 4500 pieces of equipment were examined. On 17 April 2019, Turbogenerator 4 (TG-4) of the Armenian NPP was shut down for Design Lifetime Extension and maintenance activities. On 1 June 2019, Unit 2 of the Armenian NPP was taken offline for 2019 planned outage for preventive maintenance and Design Lifetime Extension activities. On 8 September 2019, TG-3 of the Armenian NPP was put into operation, and on 25 September 2019 TG-4 of the Armenian NPP was put into operation.

At the Armenian NPP one of the most important activities of outage-2021 has started. The joint efforts of the specialists of "NVAER" company (a branch of «Atomenergoremont» JSC) engaged by «Rusatom Service» JSC, and of the Armenian NPP were directed towards the recovery of physical-mechanical properties of the reactor vessel metal. For this purpose, the annealing facility was hoisted down into the reactor vessel, and the very process of thermal treatment started, and it done approximately in 14 days. The annealing activities are implemented under the designer's supervision of the «Kurchatovsky Institute» SRC representatives.

The reactor vessel is the "heart" of the unit, and it is its lifetime that defines the service life of the plant. The annealing will securely contribute to the Unit 2 lifetime extension and its safety improvement. For the first time, the annealing technology was applied in 1987 at the third power unit of the Novovoronezh NPP. Reduction annealing has already been carried out at VVER-440 reactors of such nuclear power plants as Kola NPP (Russia), Rovno NPP (Ukraine), Kozloduy NPP (Bulgaria) and Loviisa NPP (Finland).

On 20 December 2018, the Government of Armenia approved the Agreement on the Nuclear Safety and Stress-Test Financing of the Republic of Armenia, signed between Armenia and the European Union. The total estimated cost of the project is 6.5 million euros. Based on the preliminary results of stress tests and assessment of safety margins, the safety upgrade of the ANPP has begun. At the same time, the weaknesses and areas for improvement have been identified, along with specific recommendations which the ANPP should implement to receive the ANRA's licence for Unit 2 lifetime extension.

On 15 September 2021 the Minister of Emergency Situations, the Chairman of the Nuclear Safety Committee in the Government of RA and the General Director of Armenian Nuclear Power Plant CJSC signed a memorandum of cooperation to improve data collection and exchange through the seismological system and to study the seismic mode of the ANPP and the surrounding area. This signing of the memorandum will give the opportunity to further increase the cooperation of a structure of strategic importance for the country, which is aimed at increasing the security of the Armenian nuclear power plant.

In the summer of 2005, then General Director of the IAEA M. ElBaradei visited Armenia. During this high level meeting, Armenia was assured IAEA assistance to coordinate activities on the upgrade of the ANPP.

From 5 to 6 May 2019, IAEA Director General Y. Amano visited Armenia a second time. The then Director General of the IAEA first thanked the group with a welcome and noted it was his second visit to the Armenian NPP. His first visit took place in 2012, and the team noted the large scope of activities aimed at safety enhancement and plant life extension that was implemented over the years.

Director General Amano mentioned active and ongoing IAEA technical expertise to support the effort to implement the recommendations and suggestions received as a result of IAEA missions and asked about the siting of the new NPP. The Deputy Minister of the Ministry of Territorial Administration and Infrastructure of the Republic of Armenia answered that the new NPP will likely be constructed in the area of the site of existing NPP. Mr. Amano also noted that he was assured in his meeting with the Prime Minister and the President that the Republic of Armenia would continue the peaceful use of nuclear energy in the future. In this regard, the IAEA Director General expressed his hope for close cooperation in future.

Since 1996, the Nuclear Energy Safety Council has acted under the President of Armenia. Its general duty is to report annually to the President on the state of nuclear energy safety at the ANPP. The members of the Council thoroughly review the relevant documents and appropriate specialist reports before reporting to the President. The Council consists of internationally acknowledged specialists in the field of nuclear energy. The last Council conference took place in Yerevan in August 2017.

The IAEA assembled an international team of experts at the request of the Government of Armenia to conduct an Operational Safety Review Team (OSART) review of the ANPP. Under the leadership of the IAEA, Division of Nuclear Installation Safety, the OSART team performed an in-depth operational safety review in May 2011 of the aspects essential to the safe operation of the ANPP. The conclusions of the review are based on the IAEA Safety Standards and proven good international practices. The OSART team made 14 recommendations and 12 suggestions related to areas where operational safety of the ANPP could be improved. Also, the OSART team identified good plant practices to be shared with the rest of the nuclear industry for consideration. The ANPP has already developed a detailed plan of action for implementing these recommendations and assignments, and the improvements have already started. In June 2013, an IAEA review mission visited the ANPP to revise the implementation of assignments and recommendations of the OSART mission. The mission noted the high level of the work towards fulfilling its suggestions and recommendations and, at the same time, pointed out certain issues concerning management that should be addressed. Follow-up mission recommendations and suggestions are under implementation until their full completion.

The World Association of Nuclear Operators (WANO) Moscow Centre visited to conduct a corporate peer review (CPR) of the ANPP operating organization on 20–29 May 2019. This peer review is conducted every six years, following the last CPR conducted in June 2013. The CPR team included six experts from the Russian Federation, Ukraine, Hungary, Islamic Republic of Iran and Japan, which included representation from the WANO Moscow and Tokyo Regional Centres. The WANO guiding document Performance Objectives and Criteria 2013-1 was used as a standard against which the production activities of the Armenian NPP were compared. The following seven corporate areas were reviewed as part of the mission: leadership, management, supervision and monitoring, independent supervision, support and efficiency, human resources and communication. On 23 September 2020 and at the invitation of the WANO-MC, the management of the ANPP participated in the work of the WANO-MC information club through a videoconference session. This event included a meeting with the WANO president Nikolay Sorokin, who is also the General Inspector of the Rosenergoatom Concern CJSC, as well as a meeting of the Armenian NPP and WANO Moscow Centre managements. During the session Mr. Sorokin presented the “Electric power division of the Rosatom State Corporation”. After the session a meeting was held between the ANPP and WANO MC managements to discuss issues related to the cooperation between both organizations. In addition, issues related to the current cooperation under the conditions of COVID-19 pandemic threat were discussed.

The 9th session of the Committee for the Joint Coordination of the ANPP Unit 2 Design Lifetime Extension Programme took place on 4 July 2019 in Yerevan under the co-chairmanship of the Minister of the Republic of Armenia Territorial Administration and Infrastructures and the First Deputy of the General Director of the Rosatom State Nuclear Energy Corporation. During the course of the session, the status of implementation taken during the previous session, as well as implementation of activities envisaged by the Armenian NPP Unit 2 Modernization and Operating Lifetime Extension Programme, was discussed. The meeting participants as also discussed a fresh programme of activities until 2021, extending the validity of the Intergovernmental Agreement on Cooperation between the Russian Federation and Armenia on this project. The parties also considered the preparation of a pilot project for the export and reprocessing of a batch of spent nuclear fuel.

Following the accident at the Fukushima Daiichi NPP, the interest of the public, media and the Government on issues related to nuclear safety increased significantly. In particular, the Government requested the ANRA and the ANPP to increase efforts on nuclear safety and emergency preparedness and to join the European Union initiative on conducting stress tests. In August 2015, the ANRA, based on the ANPP self-assessment report, submitted the National Report on Stress Test for Armenian Nuclear Power Plant, developed in accordance with the European Nuclear Safety Regulators Group (ENSREG) technical specifications, to the European Commission for peer review. The national report was posted on the ENSREG web site and also on the ANRA web site. The desktop review of the Armenian National Report on Stress Tests by ENSREG Members and the European Commission took place from 15 February to 31 March 2016. As a result, 196 questions on the national report from the European Commission and the ENSREG members were sent to the ANRA on 29 April 2016. The answers to the questions posed were prepared by the ANRA, the Nuclear and Radiation Safety Center and the ANPP and were provided to the European Commission on 31 May 2016. The European Commission Peer Review Mission to Armenia took place from 20 to 24 June 2016. The European Commission peer review team (PRT) was composed of 10 European Union experts (8 from European Union Member States who had been nominated by the ENSREG members and 2 from the European Commission). The conclusions of the European Commission experts to the ANPP stress test are summarized in the PRT report. Based on the results of the PRT of the ANPP stress test:

- The ANRA proposed that the ANPP perform a seismic reevaluation of systems and components with the maximum ground acceleration 0.42 g (Review Level of Earthquake) and develop a monitoring programme for volcanic activity around the ANPP.
- The seismic stability of both the foam fire extinguishing system and the water fire extinguishing system will be improved. In 2017, an analysis of the seismic stability of pipelines and equipment activities was completed. Also, the plant will be provided with design diagrams.
- The safety analysis report will be updated to consider both the implemented and scheduled activities on the ANPP modernization and safety upgrade (regarding new parameters of design basis accidents). The updated report will reflect the current status and scheduled activities of stress tests, the results of the ANPP design and actual status in compliance with requirements of standard documentation analysis and the results of the ANPP configuration control and management analysis.

Within the framework of the international assistance program implemented by the United States of America, a delegation consisting of the Government of the United States of America and PNNL representatives visited the ANPP to discuss the current activities with the ANPP management in 2019.

2.2.3 Start of the 2021 outage

On 15 May 2021 at 00:25 the ANPP Unit 2 shutdown started and at 11:45 the unit was disconnected from the RA power grid. The NPP was shut down for annual outage, which lasted 141 days owing to an unprecedented large scope of activities of the final stage of the ANPP Unit 2 life extension program.

The following main activities were performed during the 2021 outage:

1. general reactor overhaul with complete core unloading,
2. modernization of the sealing surfaces of the reactor facility upper unit,
3. reactor vessel annealing,
4. inspection of the reactor vessel metal prior to and after the annealing,
5. maintenance of the spent fuel storage pond (2SFSP) and pressurizer (2Prz),
6. general overhaul of the steam generators and eddy current testing of the steam generator tubing,
7. inspection of the composite welds of steam generators,
8. routine maintenance of the reactor coolant pumps,
9. modernization of the emergency core cooling system,
10. general overhaul of the pressurizer safety valves and steam generator safety valves,
11. routine maintenance of turbines 3 and 4,
12. intermediate maintenance of generator 3,
13. general overhaul of generator 4,
14. routine maintenance of autotransformer ??-1,
15. modernization of the in-core monitoring system,
16. replacement of instrumentation and control equipment complete sets, replacement of instrumentation cables within the boundaries of the steam generator compartment,
17. installation of additional parameters of the computer information system.

The employees of almost all the departments of the plant, as well as about 400 specialists from Belarus, Croatia, Czech Republic, the Russian Federation, Slovakia, Ukraine and other countries, who have an extensive experience of performing maintenance of NPP systems and equipment, participated in the scheduled activities of the 2021 outage. The specialists of the RA Committee for Nuclear Safety Regulation played a special role in this process.

The unit start-up was scheduled for 2 October 2021, but unit 2 connected to the grid on 17 October 2021.

In 2021 in compliance with the framework of the ANPP life extension program the capacity of Unit 2 was increased to 448 MW due to turbine improvements. Nevertheless, ANPP has the operational permission for 98 % of thermal capacity of the nuclear reactor or 407.5 MW of electrical power.

During the current year, the following main activities will be performed: processing of the reactor lid by heat treating (annealing) to increase its strength to resist neutron fluxes and replacement of the second turbogenerator.

2.2.3. Permanent shutdown and decommissioning process

A number of Government decrees were adopted with regard to decommissioning of the ANPP:

- A special fund for decommissioning of the ANPP was created under the Ministry of Finance, and the ANPP regularly makes allocations to that fund from the amount included in the ANPP electricity tariff. The ANPP Decommissioning Fund is functioning properly.
- The Management Board of the fund was created. The Deputy Prime Minister of Armenia was elected as the Chairman of the Board, which includes a number of Government members.

Under the framework of the Action Plan of the European Union Neighbourhood Policy, technical assistance was provided for developing the ANPP Decommissioning Plan as well as the Radioactive Waste Strategy. On the basis of this assistance, the Government Session No. 48 Protocol Decision, Point 8, Strategy of Armenian NPP Decommissioning, was adopted on 29 November 2007. Additional progress will mostly depend on the ANPP Decommissioning Plan.

TABLE 6. STATUS OF DECOMMISSIONING PROCESS OF NUCLEAR POWER PLANTS

Reactor unit	Shutdown reason	Decommissioning strategy	Current decommissioning phase	Current fuel management phase	Decommis-sioning licence	Licence terminated year
Armenia 1	February 1989 – other economic reasons	SAFSTOR	Reactor core defueling	Storage in an off-site facility	ANPP CJSC	—

2.3. FUTURE DEVELOPMENT OF NUCLEAR POWER SECTOR

2.3.1. Nuclear power development strategy

Armenia's energy policy is largely focused on realization of the strategy programme to provide the country with the required quantity of electric energy and gas.

In 2001–2002, an Energy and Nuclear Power Planning Study for Armenia was developed under the framework of the IAEA Programme on Technical Cooperation, and details were published in July 2004 as IAEA-TECDOC-1404. The publication included the future energy demand forecast for Armenia and the capacities which will be needed to cover that demand. During the study, two options for the development of the energy sector of Armenia were considered:

- The use of thermal power plants (TPPs) only;
- The use of both TPPs and NPPs.

It was decided that the second option for development of the energy sector was preferable, taking into account the criteria of energy safety and energy independence, as well as environmental and social considerations. Based on this study, the Least Cost Generation Plan and the Comprehensive National Energy Strategy and Energy Sector Improvement Action Plan were developed in 2006 and updated in 2014. On the basis of the aforementioned documents, the Government adopted several energy development programmes for Armenia.

On 1 November 2007, Government Decree No. 1296, The Armenian Ministry of Energy Action Programme according to the National Security Strategy, was adopted. According to this programme, it was envisaged that the new nuclear power unit(s) be put into operation immediately after the shutdown of the existing one to cover the lack of capacity. According to the document, the country's need for energy independence was considered and preference was given to the 1000 MW(e) nuclear power unit.

On 22 December 2011, the Government Session No. 50 Protocol Decision, Point 14, On Approval of the Concept for Ensuring Energy Security in the Republic of Armenia, was adopted. This document restated the importance of increasing the safety level of Unit 2, and based on the importance of national energy security and independence, the necessity of constructing a new unit. This decision also explored the possibility of continuing operations of ANPP Unit 2 after 2016.

On 23 October 2013, the President adopted the Energy Security Ensuring Concept of the Republic of Armenia, according to which Armenia would continue use of the existing nuclear unit until commissioning of a new one.

On 31 July 2014, the Government Decree No. 836-N, Measures of the Concept of the Energy Security Schedule for 2014–2020, was adopted.

On 10 December 2015, the Government Session No. 54 Protocol Decision, Point 13, Long term (until 2036) Development Pathways for the Armenian Energy Sector, was adopted. The need for nuclear development was once again stated — a measure which would ensure the necessary level of energy security and independence by 2027 through operation of a new NPP with up to 600 MW capacity.

On 14 January 2021, the Government Session No. 48-L, Armenian Energy Sector Development Strategy Program (until 2040), was adopted. The need for nuclear development was once again stated.

Several activities were carried out to support construction of a new nuclear unit.

On 27 October 2009, the Law on Construction of a New NPP in the Republic of Armenia was adopted, which serves as the legal basis for construction of a new NPP in Armenia. According to the Law on Safe Usage of Nuclear Energy in Peaceful Purposes, construction of a new NPP or decommissioning of the existing NPP are possible only after adoption of a relevant law.

The company WorleyParsons was selected in May 2009 through an international tender as a management company for the construction of the new nuclear power unit. Since its selection, WorleyParsons has finalized the development of a banking feasibility study, which is necessary for the involvement of investors.

As a result of the banking feasibility study, a Russian NPP-92 (AES-92) design (capacity — 1060 MW; operation lifetime — 60 years) was selected and approved for the nuclear island of a new NPP by Government Decision No. 1458 on 3 December 2009. The turbine island and control system of the new nuclear unit will be selected based on a tender.

The Ministry of Energy Infrastructure and Natural Resources, assisted by the United States Agency for International Development (USAID) Project for Assistance to the Energy Sector of Armenia for Energy Security and Regional Integration, developed the environmental background information report.

On 26 March 2010, Rosatom and the Ministry of Energy and Natural Resources signed an agreement on nuclear island equipment reservation aimed at equipping the new nuclear unit in Armenia.

An agreement between the Armenian and Russian governments was signed on 20 August 2010 to envisage the nuclear island equipment supply provisions and has already been ratified. Other nuclear unit components of the project (i.e. turbine island and instrumentation and control systems), are subject to negotiations with suppliers.

On 12 January 2017, Government Decision No. 122-N approved the activity and priority task plan which foresees submitting the schedule for the NPP construction to the Government of the Republic of Armenia upon approval of the plan.

The design safety requirements for NPP unit(s) were adopted by Government Decision No. 1411-N on 8 November 2012.

The Method on Seismic Hazard Assessment regarding new nuclear unit site safety requirements for NPP unit(s) was adopted by the Government in Decision No. 1546-N on 13 December 2012.

Site safety requirements for new NPP unit(s) were adopted by the Government in Decision No. 708-N on 4 July 2013.

The list of internal legal acts applied in the field of atomic energy utilization in Russian and in English was adopted by the Government in Decision No. 709-N on 4 July 2013.

However, owing to the Design Lifetime Extension of ANPP Unit 2, construction activities on a new unit have been postponed.

TABLE 7. PLANNED NUCLEAR POWER PLANT

Reactor unit/project name	Owner	Type	Capacity (MW(e))	Expected construction start year	Expected commercial year
Armenian NPP-2	Metsamorenergoatom	PWR	600–1000	TBD	—

According to the Armenian Energy Sector Development Strategy Programme (until 2040) the construction of a new medium capacity power unit which will replace the existing Unit 2 after its shutdown is being considered.

On 15 October 2021, during the meeting of the Minister of Territorial Administration and Infrastructure of the RA and the General Director of Rosatom, State Corporation discussed the programs for extending the operating life of the existing power unit of the ANPP and issues of deepening cooperation in the field of nuclear energy. Issues related to the construction of a new nuclear power unit in Armenia were also discussed.

On 29 October 2021, Deputy Prime Minister of RA stated the construction of a new NPP with small modular units with a capacity of 55 and 100 MW is being considered.

2.3.2. Project management

The Government Decree On Establishment of a Closed Joint Stock Company (CJSC) Aimed at Construction of a New NPP in the Republic of Armenia, was adopted on 3 December 2009. Metsamorenergoatom CJSC was established with the involvement of the Government and Atomstroyexport CJSC, which was delegated by Rosatom, a Russian state corporation. Metsamorenergoatom CJSC is open to other investors and has already received a licence for selecting the site for construction of the new unit.

2.3.3. Project funding

According to the intergovernmental agreement between Armenia and the Russian Federation on cooperation on construction of new nuclear unit(s) on the territory of Armenia, the Russian Federation's portion of investment will be equal to the cost of the nuclear island; the remaining funding required is expected to be covered by Armenia or other investor(s). The Government of Armenia will continue the negotiation process with potential investors for the project.

2.3.4. Electric grid development

Investigations to develop a new 400 kV network in Armenia (new voltage level in the country), as well as its expansion to neighbouring power systems, were conducted by the Energy Network Design Institute of Armenia within the framework of the Development of the Armenian Electrical Grid Scheme project (2010, 2015, 2020).

Some of the principal conclusions of this study are as follows:

- To ensure the admissible voltage level and to reduce active power losses in the electrical grid, it is desirable to construct a new 400/220 kV substation, Noravan, with the input/output of the double circuit Islamic Republic of Iran–Armenia high voltage line of 400 kV.
- It is necessary to install in the Hrazdan TPP 400 kV substation to ensure the allowable voltage levels and adjust reactive power flow.
- Calculations of short circuit currents show that there is no need to replace any equipment in existing substations or add any extraordinary additional equipment in a new 400 kV network.
- Connection of a new 400 kV overhead line and increasing electricity export to neighbouring power systems will greatly reduce the risk of unstable operation of the ANPP and the power system as a whole. The high voltage lines Armenia–Islamic Republic of Iran and Armenia–Georgia of 400 kV are currently under construction.

2.3.5. Sites

A seismic hazard assessment of the ANPP site was performed more than a decade ago, where the terms of reference for the seismic hazard assessment were developed. The seismic hazard assessment draft report was submitted to IAEA in August 2010 for expert review. Following this, the IAEA mission reviewing the seismic hazard assessment provided a number of comments and recommendations for its completion.

The Seismic Hazard Assessment for the Construction Site of a New Power Unit at the Armenian NPP — Final Report was completed in February 2011, based on the latest IAEA guidance. The report also includes a volcanic hazard assessment of the ANPP site.

The final report of IAEA experts on the final version of the seismic and volcanic hazard assessments was provided to the Armenian Government in December 2011. The second IAEA mission provided comments of an editorial nature and recommended carrying out additional investigations regarding the Yerevan fault.

In 2012 Metsamorenergoatom CJSC received the licence for selecting the site for construction of the new unit.

2.3.6. Public awareness

Public hearings on the Armenia New Nuclear Unit Environmental (Environmental Impact Assessment) Report were conducted on 17 May 2011 in Armavir and on 24 May 2011 in Gyumri. Based on the comments and recommendations made during those hearings, the report was expanded and submitted to the Ministry of Nature Protection. The Ministry of Nature Protection stated that it will provide a final conclusion after the results of design activities are incorporated in the final report.

2.4. ORGANIZATIONS INVOLVED IN CONSTRUCTION OF NPPs

Tender will be invited under the circumstances of appropriate funding, and construction firms will be identified. Armenian construction firms will be heavily involved in the construction work on the new NPP based on their capabilities.

2.5. ORGANIZATIONS INVOLVED IN OPERATION OF NPPs

The Armenian Nuclear Power Plant CJSC is the ANPP's operating organization.

Metsamorenergoatom CJSC will be the operating organization for the Armenian New Nuclear Unit.

2.6. ORGANIZATIONS INVOLVED IN DECOMMISSIONING OF NPPs

A number of Government decrees were adopted with regard to decommissioning the ANPP:

- A special fund for decommissioning the ANPP was created under the Ministry of Finance, and the ANPP regularly makes allocations to that fund from the amount included in the ANPP electricity tariff. The ANPP Decommissioning Fund is functioning properly.
- The Management Board of the fund was created. The Deputy Prime Minister of Armenia was elected Chairman of the Board.
- The Government adopted the ANPP Decommissioning Plan in November 2007.

2.7. FUEL CYCLE INCLUDING WASTE MANAGEMENT

Armenia has no nuclear fuel cycle industry and uses an open nuclear fuel cycle scheme. Until now, all nuclear fuel has been supplied by the Russian Federation. Originally, the spent nuclear fuel generated by the ANPP was reprocessed and disposed of by the agencies of the Soviet Union. Yet, since the restart of Unit 2, spent nuclear fuel has been retained on the ANPP site.

The ANPP operates with a three year fuel cycle. The spent nuclear fuel is kept in wet nuclear fuel storage in fuel ponds at the ANPP before it is transferred to dry storage.

In 2000, construction of the first stage of the spent fuel dry storage was completed. The construction was commissioned by the French firm Framatome and financed by the Government of France. The spent fuel dry storage facility was put into operation, and all the transportation of spent fuel is performed according to the requirements of the licence granted by ANRA. Now, the total volume of the first stage of storage is filled with spent fuel.

In 2005, an agreement was signed with the French company TN International S.A. for construction of three additional stages of the dry storage facility. The financing was allocated from Armenia's state budget. The second stage was completed in 2008. Since that time, a first portion of the spent nuclear fuel was transferred into dry storage. The third stage of spent fuel dry storage was constructed and is also currently in operation.

Government Session No. 43, Protocol Decision 19, The Concept for Safe Management of Radioactive Waste and Spent Nuclear Fuel in the Republic of Armenia, was adopted on 4 November 2010. According to the concept, significant activities anticipated in that area are regulated and distributed between the departments of the Government of Armenia.

The final spent fuel and high level radwaste treatment and disposal plan will be developed and included in the ANPP Decommissioning Plan.

According to the ANPP design, the annual Unit 2 radwaste generation is 308 m^3 of solid low level waste, 1.5 m^3 of solid medium level waste, 0.3 m^3 of solid high level waste and 108 m^3 of liquid medium level waste. At the ANPP, there is storage for both solid and liquid radwaste.

High level waste is stored in a special room of the reactor building. The storage area consists of 380 cells. The storage capacity is 78.34 m^3.

Medium level radwaste is stored in the Special Building. Storage capacity is 1001.22 m^3. Also, the deep evaporating facility containers are stored temporarily on the upper unheated floor of the Special Building. Its effective storage volume is 655 m^2 (3000 containers).

Liquid radwaste is stored in the Special Building. Liquid waste (evaporator residues) generated in the evaporators during drain water reprocessing are collected in the evaporator residue tank.

The storage facility for low level radwaste consists of two compartments, each measuring 27 m × 36 m × 8.9 m. The total storage volume is about 17 050 m^3.

In March 2007, the Radioactive Waste Decontamination CJSC was transferred under the Ministry of Energy and Natural Resources. Currently, medical and industrial ionizing sources are kept at the facility. Work is under way to modernize the Radioactive Waste Decontamination CJSC storage facility to also store the medium level liquid radwaste generated by the ANPP.

Within the framework of the European Union programme, ITER Consult Consortium (ITER, Sogin, Iberdrola, STUK and other organizations) provided assistance to develop the Radioactive Waste and Spent Fuel Safety Management Strategy for Armenia.

Government Session No. 42, Protocol Decision No. 50, Strategy of Safe Management of Radioactive Waste and the Spent Nuclear Fuel Generated in the Republic of Armenia, was adopted on 5 October 2017. According to this decision, the Ministry of Energy Infrastructure and Natural Resources developed an action plan and a schedule for implementing its provisions on the safe management of radioactive waste and spent fuel generated in Armenia for 2018–2026. This action plan was adopted by Government Decision N-3L on 10 January 2019.

2.8. RESEARCH AND DEVELOPMENT

2.8.1. R&D organizations

The main organizations and institutions involved in nuclear energy activities for Armenia are CJSC ANPP, CJSC Armatom, CJSC Atomservice, CJSC Nuclear and Radiation Safety Centre, CJSC Tekatomenergo and CJSC Scientific Research Institute of Energy.

2.8.2. Development of advanced nuclear power technologies

No information available.

2.8.3. International cooperation and initiatives

Armenia was elected a Member of the Board of Governors for 2017–2018 by the 61st Annual Regular Session of the IAEA General Conference in September 2017.

In 2004, Armenia joined the International Project on Innovative Nuclear Reactors and Fuel Cycles (INPRO), an IAEA initiative, in order to address the needs considering economic, safety, non-proliferation and waste management aspects of nuclear energy and its fuel cycle with innovative technology. Armenia fulfilled the collaborative project entitled Implementation Issues for the Use of Nuclear Power in Smaller Countries. This project provides small countries the opportunity to learn about problems that could arise with the construction of new nuclear units in their countries. Outcomes from the INPRO “SMALL” collaborative project were published as IAEA-TECDOC-1778. Armenia has participated in a few other collaborative projects, including SYNERGIES (Synergistic Nuclear Energy Regional Group Interactions Evaluated for Sustainability), KIND (Key Indicators for Innovative Nuclear Energy Systems), Roadmaps for a Transition to Globally Sustainable Nuclear Energy Systems and the TNPP-2 project relating to transportable NPPs. Results of the collaborative SYNERGY, KIND and ROADMAP projects were already published as IAEA TecDocs. In 2020, National Polytechnic University of Armenia (NPUA) took part in the technical cooperation project INMA Master's Programmes in Nuclear Technology Management (published in NG-T-6.12).

Armenia was invited to join the Global Nuclear Energy Partnership (GNEP). On 1 October 2008 the agreement was signed and Armenia became a member of the GNEP, which provides significant benefits to Armenia’s nuclear programme (a change of name to the International Framework for Nuclear Energy Cooperation was adopted in June 2010).

Armenia maintains bilateral cooperation, mostly concerning safety of the ANPP, with countries such as Argentina, France, Italy, the Russian Federation, the United Kingdom and the United States of America. Armenia also participates in several international projects developed in the framework of cooperation under the aegis of the IAEA, European Commission and USAID.

Armenia maintains close cooperation with the IAEA in pursuit of nuclear power. Armenia became a Member State of this organization in 1993, and IAEA experts have participated in many assistance projects since then. In April of 1993, the Government of Armenia made the decision to restart Unit 2 of the ANPP, and IAEA experts participated actively in pre-commissioning investigations and evaluation of the condition of plant equipment. Moreover, they elaborated the whole concept of Unit 2 recommissioning. Armenia is also collaborating with the IAEA in the field of upgrading nuclear safety. At present, several national programmes of the ANPP Unit 2 safety upgrade are in different phases of implementation. The IAEA is continuously assisting the ANRA, providing appropriate support and periodic recommendations.

Since 1996, the United States Department of Energy and the European Commission have provided technical assistance in upgrading the level of ANPP operations, as well as modernizing technological equipment in the plant.

Over the years, other countries such as the Czech Republic, France, Italy, Russian Federation (since 2008) and the United Kingdom, have joined assistance programmes and related efforts.

Armenia cooperates with Argentina within the bilateral project Creation in Armenia of a Center for Training and Qualification in Non-Destructive Metal Testing Techniques with the assistance of the IAEA.

There are many joint projects with the Russian Federation within the framework of the Nuclear Safety Assistance Programme. In 1996, an agreement was signed between the ANPP and Rosenergoatom on industrial and technical-scientific cooperation. In 2000, an agreement was signed between the governments of Armenia and the Russian Federation on Cooperation in the Field of Peaceful Use of Nuclear Energy.

In the framework of bilateral cooperation between Armenia and the United States of America in 2001, within the Armatom Institute, the International Nuclear Safety Center of Armenia was created. The Joint Statement on Cooperation between International Nuclear Safety Centers of Armenia and the United States of America was signed on 7 February 2001.

Since 1996, the ANPP has been a member of WANO. WANO Moscow Centre has commissioned two inspections relevant to ANPP operational safety.

The ANRA has cooperation agreements with the nuclear regulatory authorities of the following countries: Argentina, the Russian Federation, Ukraine and the United States of America. The ANRA is a member of the Forum of the State Nuclear Safety Authorities of the Countries Operating WWER Type Reactors. The ANRA also participates in the CONCERT (Concertation on European Regulatory Tasks) group.

In 2007, the Government of Armenia made the decision to join the agreement between the governments of Kazakhstan and the Russian Federation on the establishment of the International Uranium Enrichment Centre in Angarsk.

By Government Decree No. 1407A of 27 August 2020 the proposal to sign the 2020 / 042-470 financing agreement between the European Commission and the Republic of Armenia on "Promotion of Nuclear Safety Culture. Armenia (Component B)" is approved.

By Government Decree No. 985A of 17 June of 2021 the proposal to sign the 2020 / 042-470 financing agreement between the European Commission and the Republic of Armenia on "Promotion of Nuclear Safety Culture. Armenia (Component A)" is approved.

2.9. HUMAN RESOURCES DEVELOPMENT

In view of energy security and energy independence, Armenia gives special attention to the development of nuclear energy in the country.

Activities towards the construction of a new nuclear unit in Armenia began in 2008. A Law on Construction of a New NPP in the Republic of Armenia was adopted on 27 October 2009 and serves as the legal basis for construction of a new NPP in Armenia.

The need for qualified specialists is growing in importance for Armenia with regard to construction of new nuclear units as well as for operation, implementation of continuous safety improvements and decommissioning of the ANPP.

Armenia is the only country in the entire Caucasus region that has operated an NPP for over 30 years. Qualified specialists are required for the already existing ANPP, the ANRA, Nuclear and Radiation Safety Centre, Armatom and other research institutes to address issues and challenges in view of new developments in nuclear energy in Armenia.

Armenia has two main institutions preparing nuclear experts: NPUA and Yerevan State University (YSU). Armenian specialists from ANRA, the NPP and support organizations participate in scientific visits and training in Europe, the United States of America and other areas, conducted under the auspices of IAEA technical cooperation projects and international aid programmes.

To increase the quality of nuclear specialists, two departments of YSU and the NPUA currently provide specialized education in the field of nuclear energy. However, enhancing the integrated education system for the nuclear sector in Armenia is essential. Therefore, the Government approved a concept on human resources management. Implementation of knowledge management for all phases, including design, construction and commissioning, operation and decommissioning, for both existing and future NPP units are the main parts of the concept.

An evaluation of human resource needs in conjunction with the new NPP in Armenia was conducted under IAEA Technical Cooperation Project ARM-005. The report of that feasibility study of nuclear energy development in Armenia, titled Evaluation of Human Resource Needs in Conjunction with New NPP Build, was completed in 2008 and published as IAEA-TECDOC-1656 in 2011. The analysis, which covers all stages of construction of the new nuclear power unit, relates both to the sponsoring organization and to the regulatory agency dealing with nuclear power in Armenia.

Armenia is currently engaged in the following activity: NPUA and YSU closely cooperate with the International Nuclear Management Academy (INMA) in the framework of the IAEA and will open a Nuclear Technology Management (NTM) Master study programme.

Armenia has expressed a strong interest in the International Nuclear Management Academy (INMA) programme and requested that the IAEA conduct an INMA Initial Assist Mission. The mission took place from 11 to 14 June 2018 at NPUA to assess the feasibility of implementing a Master degree programme to meet INMA requirements and to establish organizational support and a network between the university and relevant stakeholders. The goals of the Initial Assist Mission were achieved. It was confirmed that elements of NPUA's and YSU's existing programmes are closely related to some aspects of a possible INMA nuclear technology management programme which the universities jointly intend to introduce. The INMA Final Peer Review Assessment Mission took place from 21 to 24 October 2018 at NPUA. In October 2019, significant progress was observed that will provide a solid foundation as the programme is further developed.

On 14 May 2020 by order of Minister of Education, Science, Culture and Sport, the Nuclear Technology Management Master study programme has been included on the list of education specialties of Republic of Armenia. NPUA now submits corresponding documents to the Ministry of Education, Science, Culture and Sport for license for the Nuclear Technology Management Master study programme. By Order of Ministry of Education, Science, Culture and Sport of the Republic of Armenia on 06 October 2020, NPUA received license for the implementation of 041301.35.7 "Nuclear Technology Management" Study Programme for full-time and part-time students and the first students enrolled in the "Nuclear Technology Management" Study Programme in September 2021.

Significant expansion of staffing at the former Ministry of Energy Infrastructure and Natural Resources and CJSC Metsamorenergoatom to support new unit design and procurement is expected after selection of strategic partners and investors. Armenia's contract with WorleyParsons requires development of specific training plans for personnel working on the preconstruction and construction phases of the project and for personnel responsible for project safety.

Enhancement of the Armenian nuclear educational system and comprehensive development and upgrade of the training system for personnel within the nuclear power sector will include the development and upgrade of the following aspects of the training system:

- Management of training system development and operation;
- Organizational structure and staffing of the training system;
- The training centre;
- Training programmes and material using the systematic approach to training for various categories of personnel;
- Simulators (full scope, compact);
- Multi-functional multimedia computer based training systems for various jobs and activities;
- Training and development of instructors;
- Training and development of nuclear power sector managers.

2.10. STAKEHOLDER INVOLVEMENT

In Armenia, information support activities are performed within the framework of stakeholder communication plans, which deal with continuous provision of information related to the development of the nuclear power programme to the public.

2.11. EMERGENCY PREPAREDNESS

On 8 December 2005, an amendment was made to the Law on Population Protection in Emergency Situations, according to which, in the case of a nuclear or radiation emergency at the NPP, the functions of all involved responsible organizations shall be determined by Government decree. On 22 December 2005, Government Decree No. 2328, National Plan for Population Protection in case of Nuclear and/or Radiation Emergency at the Armenian NPP, was issued. As a result of the exercises on a nuclear or radiation emergency at the nuclear power plant conducted for checking the real possibilities to use that decree, a new edition of the National Plan for Population Protection in case of Nuclear and/or Radiation Emergency at the Armenian NPP was created, and this was adopted by Government Decree No. 194 on 17 January 2008.

3. NATIONAL LAWS AND REGULATIONS

3.1. REGULATORY FRAMEWORK

3.1.1. Regulatory authority(s)

The state authority for supervision of nuclear and radiation safety was established by Government Decree No. 573 on 16 November 1993 and is known as the State Department for Supervision on Nuclear and Radiation Safety of Utilization of Nuclear Energy for the Government of Armenia. By the same decree, the Department statute was approved and the authority was charged with the functions of inspections.

Government Decree No. 70 of 19 February 2000 authorized the Department to also have regulating functions, and, according to that decree, it prepared a new statute which was approved by Government Decree No. 385 of 22 June 2000. The Department was given a new name, the Armenian Nuclear Regulatory Authority (ANRA), in accordance with that decree. The ANRA was under direct subordination to the Armenian Government, independent from organizations responsible for the development and utilization of atomic energy. According to its new statute, the ANRA was to organize and perform state supervision and inspections over utilization of nuclear energy, as well as oversee its regulation.

On 24 May 2001, according to Government Decree No. 452, the ANRA was awarded with the authorization of state regulation of protection against irradiation from ionizing radiation sources and of their safety.

The status of the ANRA was changed again on 27 June 2002, according to Government Decree No. 912, in order to respond to the reforming principles implemented into the Armenian system of Government management. The ANRA was included in the Ministry of Environmental Protection.

On 26 December 2002, the new statute of the ANRA was approved by Government Decree No. 2183. The ANRA was renamed the Inspectorate for State Supervision on Nuclear and Radiation Safety of Utilization of Nuclear Energy under the Ministry of Environmental Protection. According to the new statute, the ANRA was authorized with the following key duties: to perform state regulation within the field of nuclear energy utilization with the main objective of securing the protection of the population, the personnel involved in the nuclear industry and the environment.

In accordance with the Ordinance of the President of Armenia adopted on 20 May 2008, the ANRA was reorganized into the State Committee on Nuclear Safety Regulation under the Government of Armenia. Now, the ANRA's task is the state regulation of atomic energy utilization (safety of nuclear facilities, safe use of ionizing radiation sources, safe management of radioactive waste, and safe transport of radioactive and nuclear materials) aimed at ensuring the safety of the population, personnel and the environment, and to defend the safety interests of Armenia.

The regulatory authority for nuclear safety is the ANRA. The licensee is responsible for the safety of the NPP. The licensee is obliged to:

- Guarantee respect of the principles, criteria and requirements on nuclear and radiation safety, as well as the conditions or acts of the temporary operation permit;
- Inform the ANRA of deviations from the conditions of the temporary operation permit, as well as of any incidents or emergencies during NPP operation.

On 25 April 2001, the Science Research Centre of Nuclear and Radiation Safety was established at the ANRA, according to Government Decree No. 342, with the aim of enabling the ANRA to carry out an independent expertise activity.

On the basis of Government Decree No. 389 of 22 August 1994 On Nuclear Power Plant Safety Norms and Rules, all the rules and norms applicable to nuclear power in the Russian Federation were accepted in Armenia.

According to Government Decree No. 252 of 7 April 2007 On Abrogation of the Government Decree No. 389 of 22 August 1994 and Item 2 of the Government Decree No. 239 of 20 April 1999, Government Decrees No. 389 of 22 August 1994, On Nuclear Power Plant Safety Norms and Rules, and No. 239 of 7 April 2007, On the List of Normative Decisions Adopted by the Council of Ministers of the Armenian Soviet Socialistic Republic and Effective Before 23 August 1990, became ineffective. Appropriate governmental bodies in Armenia are in the process of developing internal norms and standards for the nuclear sector.

Numerous IAEA inspection and assessment missions have confirmed Armenia's commitment to its international obligations, as well as a high level of openness and transparency. No violation or deviation from the requirements of international treaty was reported. Armenia fully complies with its international obligations under UN Security Council Resolution 1540. Armenia strictly follows the National Action Plan for 2015–2020 and has outlined a number of clear steps in this direction. Armenia is also actively involved in nuclear non-proliferation initiatives, such as the Global Initiative on Nuclear Terrorism and the Proliferation Security Initiative. The Armenian Government is constantly working with its international partners at the bilateral and multilateral levels to further strengthen its national capacity to combat nuclear smuggling.

3.1.2. Licensing process

The licensing process in the nuclear field is regulated by the Law on Licensing and the relevant decisions of the Government of Armenia.

3.2. MAIN NATIONAL LAWS AND REGULATIONS IN NUCLEAR POWER

The following laws concerning the activities in the field of nuclear energy use are in use in Armenia:

- The Law on Energy of the Republic of Armenia, which entered into force on 7 March 2001, and replaced the previous law, which replaced the Law on Energy of the Republic of Armenia, which entered into force on 1 July 1997.
- The Law on Safe Use of Nuclear Energy for Peaceful Purposes, which entered into force on 1 February 1999. Some amendments and additions were implemented in this law.
- The Law on Licensing, which entered into force on 27 June 2001.
- The new Law on the Export Control for the Goods of Dual Purpose and Technologies and their Transit Transportation through the Territory of Armenia, which entered into force in April 2010 and replaced the previous law on the same topic.

- Law on Implementation of Modifications and Additions in the Code of Armenia on Criminal Legal Violations, which entered into force on 1 August 2003.
- Government Decree No. 1219-N of 18 August 2006 on Radiation Safety Norms.
- Government Decree No. 1489-N of 18 August 2006 on Radiation Safety Rules.

The above mentioned laws, as well as Government decrees and all other legislative and regulative documents are presented on the official web sites of the National Assembly of Armenia (www.parliament.am), the Government of Armenia (www.gov.am), Ministry of Energy and Natural Resources (www.minenergy.am) and the ANRA (www.anra.am).

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APPENDIX 1: INTERNATIONAL, MULTILATERAL AND BILATERAL AGREEMENTS

AGREEMENTS WITH THE IAEA		
NPT related agreement INFCIRC No. 455	Entry into force:	5 May 1994
Additional protocol GOV/2948	Entry into force:	September 2004
Improved procedures for designation of safeguards inspectors		No reply
Supplementary agreement on provision of technical assistance by the IAEA	Entry into force:	30 September 1999
Agreement on privileges and immunities		Non-party
RELEVANT INTERNATIONAL TREATIES OR AGREEMENTS		
Treaty on the Non-Proliferation of Nuclear Weapons (NPT)	Acceded:	15 July 1993
Convention on the Physical Protection of Nuclear Material	Entry into force:	23 September 1993
Convention on Early Notification of a Nuclear Accident	Entry into force:	24 September 1993
Convention on Assistance in Case of a Nuclear Accident or Radiological Emergency	Entry into force:	24 September 1993
Vienna Convention on Civil Liability for Nuclear Damage	Entry into force:	24 November 1993
Joint Protocol		Non-party
Protocol to Amend the Vienna Convention on Civil Liability for Nuclear Damage		Not signed
Convention on Supplementary Compensation for Nuclear Damage		Not signed
Convention on Nuclear Safety	Entry into force:	20 December 1998
Zangger Committee		Non-member
Nuclear export guidelines		Not adopted
Acceptance of NUSS codes		No reply
Comprehensive Nuclear-Test-Ban Treaty		1 October 1996
Amendment to the Convention on the Physical Protection of Nuclear Material	Ratification:	18 March 2013
Joint Convention on the Safety of Spent Fuel Management and on the Safety of Radioactive Waste Management	Entry into force:	20 August 2013
BILATERAL AGREEMENTS		
Agreement with the Russian Federation on Restarting Operation of the ANPP	Entry into force:	17 March 1994

Agreement with the Republic of Argentina on cooperation for the peaceful uses of nuclear energy	Entry into force:	22 April 1999
Agreement with the Government of the Russian Federation on cooperation in the field of peaceful use of nuclear energy	Entry into force:	10 January 2001
Memorandum of understanding between the Government of the Republic of Armenia and the Government of the United States of America concerning cooperation in the energy sector, including renewable and nuclear energy	Entry into force:	18 October 2012
Intergovernmental agreement between Armenia and the Russian Federation on nuclear safety	Entry into force:	15 April 2014
The agreement between Armenia and the Russian Federation on early notification of nuclear accident and exchange of information in the field of nuclear and radiation safety	Entry into force:	7 October 2015
The agreement on cooperation in the field of nuclear energy between Armenia and Belarus		19 February 2016

APPENDIX 2: MAIN ORGANIZATIONS, INSTITUTIONS AND COMPANIES INVOLVED IN NUCLEAR POWER RELATED ACTIVITIES

NATIONAL ATOMIC ENERGY AUTHORITIES

Ministry of Territorial Administration and Infrastructure Government House 3, Republic Square 0010 Yerevan, Republic of Armenia	Tel.: +374 11 51-13-62 Fax: +374 11 52 63 65 http://www.mtad.am
State Committee under the Government of Armenia on Nuclear Safety Regulation (ANRA) 4, Tigran Mets Ave. 0010 Yerevan, Republic of Armenia	Tel.: +374 10 54 39 95 Fax: +374 10 58 19 62 info@anra.am
MAIN POWER UTILITY	
Armenian Nuclear Power Plant Metsamor, 377766 Armavir Region 6 Republic of Armenia	Tel.: +374 10 28 18 80 Fax: +374 10 28 85 80 anpp@anpp.am
ENERGY RESEARCH INSTITUTES, UNIVERSITIES AND OTHER ORGANIZATIONS	
National Polytechnic University of Armenia 105, Vahan Teryan St. 0009 Yerevan, Republic of Armenia	Tel.: +374 10 520 521 Fax: +374 10 520 521 a.gevorgyan@polytechnic.am www.polytechnic.am
Yerevan State University	www.ysu.am
National Academy of Sciences of Armenia	www.sci.am
Yerevan Physics Institute	www.yerphi.am
Scientific Research Institute of Energy 5/1 Myasnikyan Ave. 0025 Yerevan, Republic of Armenia	Tel./Fax: +374 10 55 96 59 official@energinst.am
MANUFACTURERS AND SERVICES	
Armatom 50, Admiral Isakov Ave. 0114 Yerevan, Republic of Armenia	Tel.: +374 10 73 46 22 Fax: +374 10 74 21 30 vpetros@@web.am
Atomservice Metsamor, 377766 Armavir Region 6 Republic of Armenia	Tel./Fax: +374 10 28 55 32 atomserviceojs@gmail.com
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